

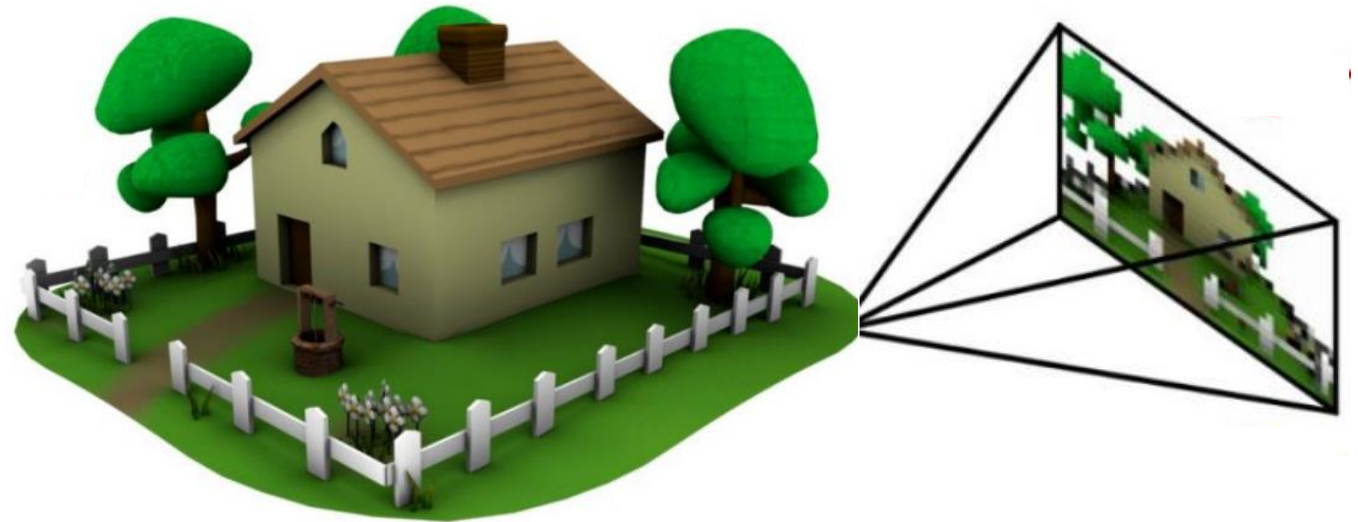
Introduction

Computer Vision

Goal of Computer Vision

extract semantic information from digital image data
to be used for decision making support or automated systems

challenging problem:
images are only 2D projections
of the 3D world



nowadays heavily powered by artificial intelligence (AI), especially machine learning (ML)

Applications of Computer Vision

facial recognition



automated inspection



autonomous driving



medical imaging



optical character recognition



augmented reality

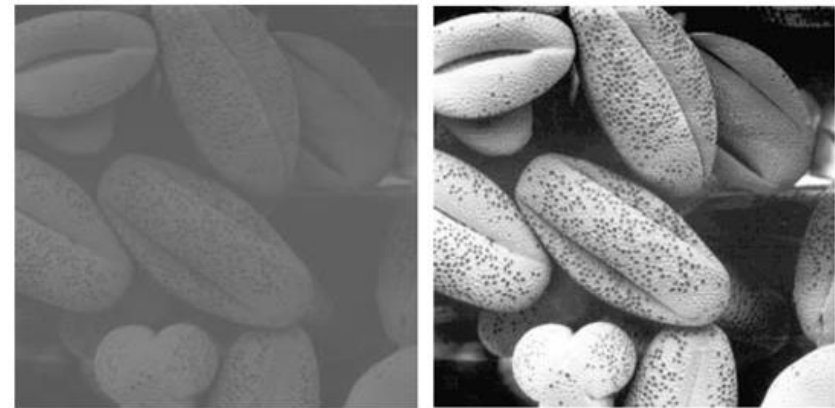
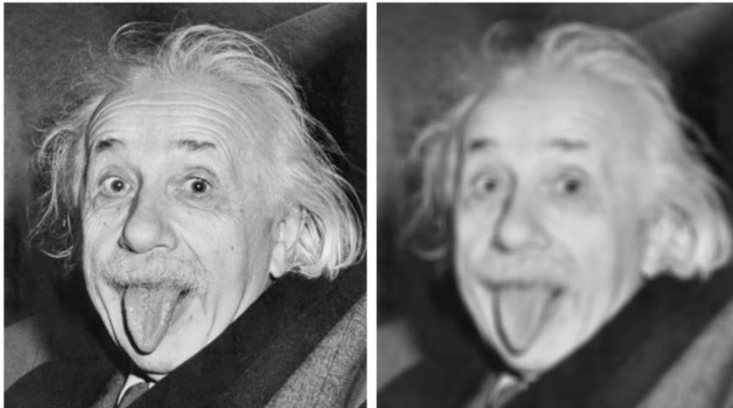


and many more ...

Image Processing

transformations from image to image

(such as scaling, smoothing, sharpening, or contrast stretching)

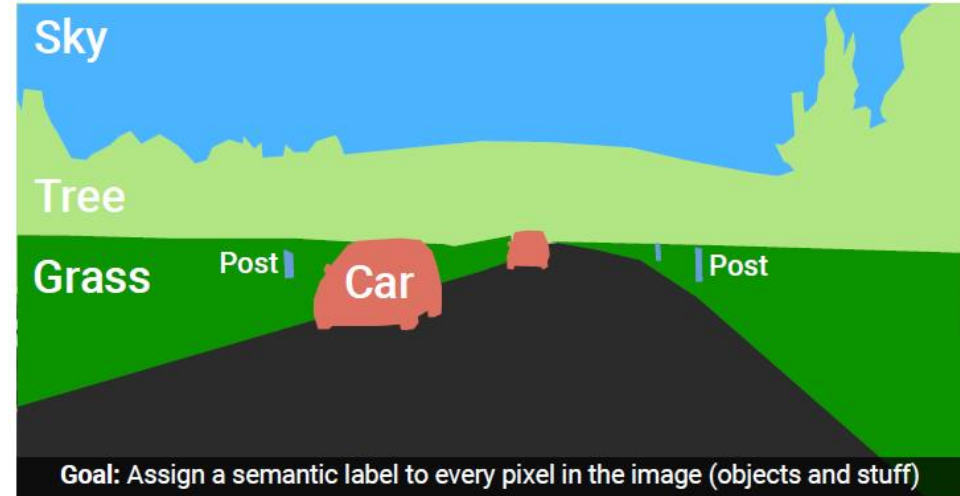


to facilitate either machine perception or just human interpretation

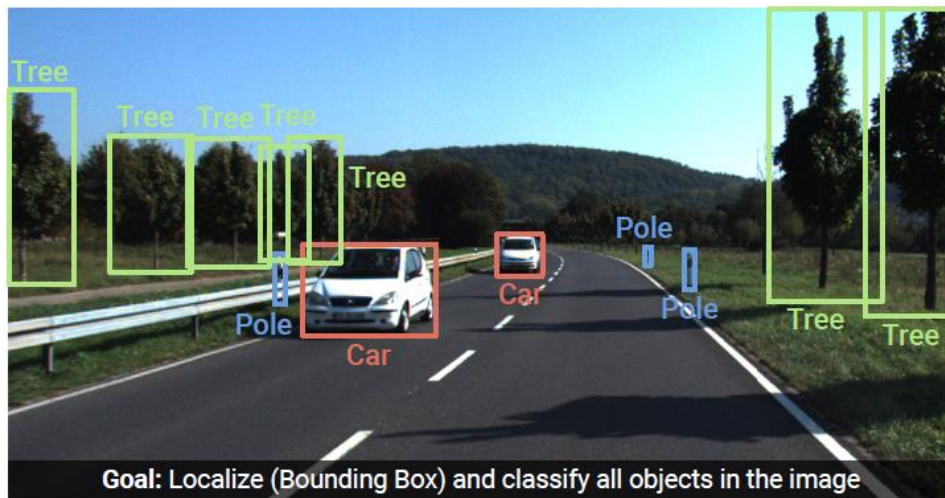
Image Understanding (Recognition)



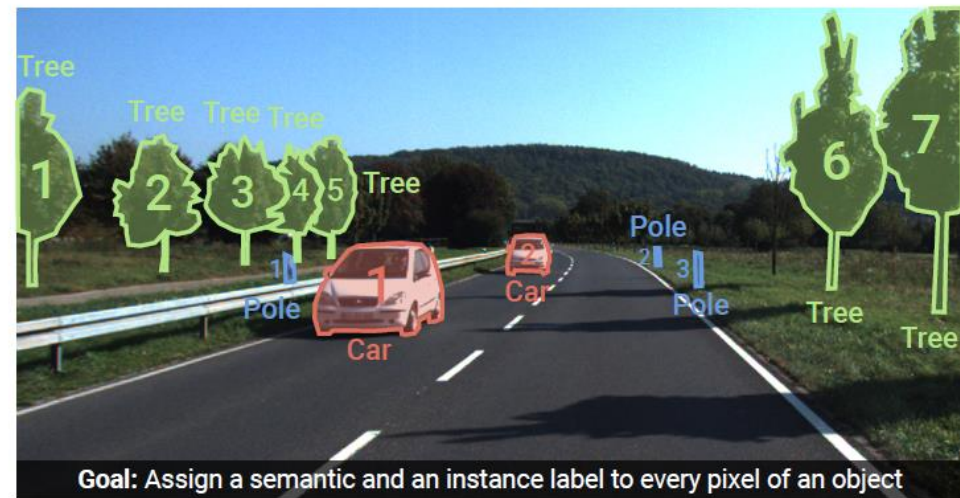
Image Classification



Semantic Segmentation



Object Detection



Instance Segmentation

Course Schedule

1. Digital Image Processing

image formation, intensity transformations, compression, spatial filtering, Fourier transform, aliasing, image pyramids

2. Edges and Features

Canny & Marr-Hildreth edge detectors, Hough transform, feature detection (Harris corner detector) and description (SIFT), feature matching, eigenfaces

3. Projective Geometry

geometric transformations, image alignment, stereo vision, camera calibration

4. Image Classification

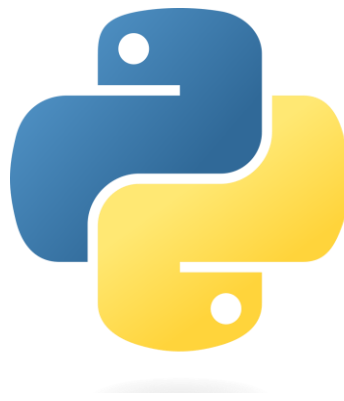
supervised learning, linear regression, bias-variance tradeoff, neural networks, convolutional neural networks, transfer learning, transformers (language models, ViT, DINO, CLIP)

5. Segmentation and Detection

semantic segmentation, object detection (R-CNN, YOLO), object tracking, instance segmentation, promptable segmentation

6. Image Synthesis

generative models, GAN, VAE, diffusion



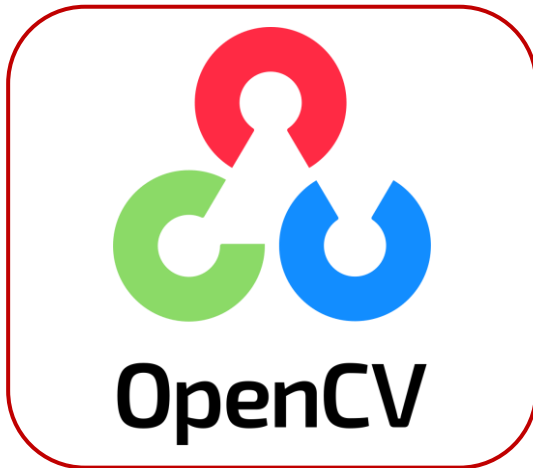
Used Programming Language: Python

- good compromise between rapid prototyping and production
- vast ecosystem
- very popular for data processing and ML: scientific Python stack



Used Python Libraries

computer vision:



deep learning:



Programming Environments

locally, best use

- a virtual environment to flexibly install packages (e.g., [venv](#))
- an IDE of your choice (e.g., [VS Code](#))
- both plain Python files or [Jupyter notebooks](#) are fine

but cloud-based environments also fine (e.g., [Google Colab](#))

Literature

- [Computer Vision: Algorithms and Applications](#)
- Digital Image Processing, Gonzales & Woods
- [Deep Learning](#)
- [The Little Book of Deep Learning](#)
- [Understanding Deep Learning](#)